

HealthConnect Clinic - Project Summary

1. Problem

HealthConnect Clinic is experiencing a significant number of missed appointments. The Week 5 objective was to explore appointment and patient characteristics, develop key performance indicators and identify patterns associated with no-shows and attendance.

2. Work Completed

During Week 5, I:

- Explored the HealthConnect appointment dataset containing 5,000 records across 18 variables.
- Reviewed appointment outcomes and patient characteristics.
- Analysed previous no-show behaviour.
- Analysed booking lead time.
- Examined reminder status and reminder channels.
- Analysed distance to the clinic and waiting time.
- Developed five key performance indicators.
- Built an interactive Power BI dashboard containing an Executive Overview and No-Show Drivers page.
- Identified key business insights and developed corresponding recommendations.

3. Key Findings

- The overall no-show rate was 51.2%, compared with an attendance rate of 48.8% among non-cancelled appointments.
- Patients with 2+ previous no-shows recorded a no-show rate of approximately 63.5%.
- Appointments booked 31+ days in advance recorded a no-show rate of approximately 63.9%, compared with 26.6% for appointments booked 0–3 days ahead.
- Appointments associated with reminders had an attendance rate of approximately 50.1%, compared with 45.4% for appointments without reminders.
- Patients travelling 21+ km recorded a no-show rate of approximately 60.5%, the highest among the distance groups with recorded distance values.

4. Recommendations

Based on the findings, HealthConnect should consider:

1. Targeted confirmation for patients with previous no-show history.
2. Additional confirmation strategies for appointments booked far in advance.
3. Reviewing reminder coverage and reminder processes.
4. Investigating accessibility challenges for patients travelling longer distances.
5. Using previous appointment behaviour as an operational indicator for prioritising follow-up.

5. Key Considerations

The analysis should be interpreted with consideration for missing values, differences in group sizes, date consistency concerns and the distinction between association and causation.

Waiting time should also be interpreted cautiously because it may reflect information available after an appointment rather than a factor known before a patient decides whether to attend.

6. Week 6 Focus

The next stage will focus on refining the analytical findings, validating the most important patterns, strengthening the dashboard and developing more actionable recommendations based on the evidence identified during Week 5.